Exact Functional Graphs of Finite Linear Maps from Rational Canonical Data

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Abstract

For every linear endomorphism of a finite-dimensional vector space over a finite field, we derive its complete functional-graph census from its invariant factors. The result simultaneously covers nilpotent, nonsemisimple, and inseparable cases. It gives every fixed count, the periodic subspace and maximal preperiod, exact periods, the Artin–Mazur zeta function, and the characteristic polynomial of composition on the full function space. Exact enumeration, independent rank calculations, and symbolic controls verify eight structurally distinct witnesses, including genuine $\mathbb{F}_4$ arithmetic.

1 Invariant-factor fixed counts

Let $A \in \mathrm{End}_{\mathbb{F}_q}(V)$ have invariant factors $f_1 \mid \cdots \mid f_r$, so that $V \simeq \bigoplus_i \mathbb{F}_q[X]/(f_i)$ and A acts as multiplication by X. Put

$$F_n = q^{\sum_i \deg \gcd(f_i, X^n - 1)}.$$

Theorem 1 (complete finite dynamics)**.** *For every $n \geq 1$, $|\mathrm{Fix}(A^n)| = F_n$. Write $f_i = X^{e_i} g_i$ with $g_i(0) \neq 0$. Then $\mathrm{Per}(A)$ is a vector subspace of dimension $\sum_i \deg g_i$, and the largest preperiod is $\max_i e_i$. The numbers of least-period points and cycles are*

$$P_n = \sum_{d \mid n} \mu(n/d) F_d, \qquad C_n = P_n / n,$$

and $\zeta_A(z) = \prod_{n \geq 1} (1 - z^n)^{-C_n}$.

Proof. On $\mathbb{F}_q[X]/(f_i)$, the kernel of multiplication by $X^n - 1$ has dimension $\deg \gcd(f_i, X^n - 1)$. Direct sums prove the fixed formula; no square-free hypothesis on $X^n - 1$ is used. Primary decomposition separates the nilpotent X-part, of height e_i, from the invertible g_i-part. This proves the periodic-subspace and tail assertions. Möbius inversion gives P_n, orbit division gives C_n, and exponentiating $\sum_{n \geq 1} F_n z^n / n$ gives the zeta product. □

2 Transient and inseparable structure

The theorem distinguishes two phenomena sometimes obscured by fixed-count tables. A factor X^e contributes no nonzero periodic coordinate but fixes the exact maximal tail e. Conversely, repeated factors coprime to X can produce nonsemisimple periodic dynamics. For example, $(X+1)^4$ over $\mathbb{F}_2$ is inseparable and has restriction order four, while $(X+\alpha)^2$ over $\mathbb{F}_4$, with $\alpha^2 + \alpha + 1 = 0$, has order six. The latter uses field arithmetic, not arithmetic in $\mathbb{Z}/4\mathbb{Z}$.

Lemma 1. *The theorem remains valid when the characteristic divides n and $X^n - 1$ is inseparable.*

Proof. The cyclic-module kernel calculation uses the ideal generated by $\gcd(f_i, X^n - 1)$ with its full multiplicity. It never replaces the gcd by a set of distinct roots. □

Source note

Historical finite-linear context appears in Elspas [1] and Xu–Zou [2]. We make no priority claim.

References

[1] B. Elspas, The theory of autonomous linear sequential networks, *IRE Trans. Circuit Theory* 6 (1959), 45–60, doi:10.1109/TCT.1959.1086506.

[2] G. Xu and Y. M. Zou, Linear dynamical systems over finite rings, *J. Algebra* 321 (2009), 2149–2155, doi:10.1016/j.jalgebra.2008.09.029.

Declarations

Scope. The exact registered literal is `NO_BAD_EULER_OR_ROOT_NUMBER`. **Data and code.** All ledgers, independent checks, and build instructions are supplied in the HCS-C204 release package. **Competing interests.** None declared. **AI-use disclosure.** Generative AI tools assisted drafting and code generation. Every theorem statement, source record, exact output, and scope claim was checked through the released internal artifact audit chain; this is not external peer review or an independent error process.